

# Sun Woo Kim

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**Email:** ska503@sfu.ca    **Homepage:** <https://pulsekim.github.io/>    **Linkedin**

## Research Interests

- Physics & kinematics based character & robot animation
- Deep reinforcement learning for character and robot motion control
- Text-to-motion retrieval and generation
- Body gesture generation for conversational agents

## Education

**Simon Fraser University** British Columbia, Canada  
Ph.D. Student in Computing Science Sep 2025 – Present  
Supervisor: Xue Bin (Jason) Peng

**Seoul National University** Seoul, Korea  
Master's degree in Computer Science Sep 2019 – Sep 2022  
Supervisor: Jehee Lee  
Thesis: *Developing Motion Control of Different Morphology using Deep Reinforcement Learning*

**Pohang University of Science and Technology (POSTECH)** Pohang, Korea  
Bachelor's degree in Creative IT Engineering Feb 2015 – Feb 2019  
Magna cum laude GPA: 3.88 / 4.3

## Work Experience

**NC AI (Spun offed from NCSOFT)** Sunnam, Korea  
Graphics AI Researcher, Motion Service Lab Feb 2025 – Sep 2025  
Research Topic: Text to motion retrieval & generation

**NCSOFT** Sunnam, Korea  
Graphics AI Researcher, Graphics AI Lab Aug 2022 – Feb 2025  
Research Topic: Gesture generation & Video to Motion

## Publications

**DSD: Learning Diverse and Reusable Motor Skills via Diffusion Skill Discovery**

**Sunwoo Kim**, Xue Bin Peng  
*arXiv preprint arXiv:2609.17682, 2026*

**Controllable Single Motion Synthesis**

Taehyun Byun, **Sunwoo Kim**, Minwook Chang, Sungjoon Choi  
*Pattern Recognition(Under Revision), 2026*

**Body Gesture Generation for Multimodal Conversational Agents**

**Sunwoo Kim**, Minwook Chang, Yoonhee Kim, Jehee Lee  
*SIGGRAPH Asia, 2024.*

## **Towards Natural Prosthetic Hand Gestures: A Common-Rig and Diffusion Inpainting Pipeline**

Seungyup Ka, Taemoon Jeong, **Sunwoo Kim**, Sankalp Yamsani, Joohyung Kim, Sungjoon Choi

*IEEE Engineering in Medicine and Biology Society (EMBC), 2024.*

## **Human Motion Control of Quadrupedal Robots using Deep Reinforcement Learning**

**Sunwoo Kim**, Maksim Sorokin, Jehee Lee, Sehoon Ha

*Robotics: Science and Systems (RSS), 2022.*

*Emerging Technologies, SIGGRAPH Asia, 2022.*

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|-----------------------|-----------------------------------------------------------------------------|------|
| Awards & Scholarships | <b>Special Grad Entrance Scholarship</b> (CS Department, SFU)               | 2025 |
|                       | <b>Runner-up: KCGS Master's Award</b> (Korea Computer Graphics Society)     | 2022 |
|                       | <b>Grand Prize: Creative ICT Forum</b> (Ministry of Science and ICT, Korea) | 2018 |
|                       | <b>Grand prize: Creative Design III</b> (POSTECH)                           | 2017 |
|                       | <b>Grand prize: Creative Design II</b> (POSTECH)                            | 2016 |

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|---------------------|---------------------------------------------------------------------|-------------|
| Teaching Experience | <b>Teaching assistant</b> , Seoul National University               | Spring 2021 |
|                     | Core Computing: Computation thinking (Prof. Jehee Lee)              |             |
|                     | <b>Undergraduate tutor</b> , POSTECH                                | Spring 2018 |
|                     | Intro to Digital System & Microprocessor Design (Prof. Jaejoon Kim) |             |

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| Professional Service | <b>Paper Reviewer</b><br>SIGGRAPH, SIGGRAPH Asia, TVCG |
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| Other Experience | <b>Booth Staff</b> , NC AI exhibition booth at SIGGRAPH<br>Presenting motion retrieval & generation research. | 2025 |
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| References | <b>Prof. Xue Bin (Jason) Peng</b><br>Assistant Professor, Simon Fraser University<br>Research Scientist, NVIDIA<br>xbpeng@sfu.ca |
|            | <b>Prof. Jehee Lee</b><br>Professor, Seoul National University<br>jeheel@snu.ac.kr                                               |
|            | <b>Prof. Sehoon Ha</b><br>Assistant Professor, Georgia Institute of Technology<br>sehoonha@gatech.edu                            |